

Student Laboratory Exercise: Wireshark Display Filters and Network Traffic Analysis

Objective

The purpose of this laboratory exercise is to learn and apply Wireshark display filters for analyzing captured network traffic. Students will generate different types of network traffic using Nmap and web browsing activities, capture the traffic using Wireshark, and use display filters to isolate and analyze specific packets and protocols.

Learning Outcomes

Upon successful completion of this lab, students will be able to:

- Capture network traffic using Wireshark.
- Generate various network scanning activities using Nmap.
- Differentiate between capture filters and display filters.
- Apply protocol, IP address, port-based, and packet-field display filters.
- Analyze TCP flags, DNS traffic, and HTTP communications.
- Investigate network scanning patterns and web traffic behavior.

Software Requirements

- Windows Operating System
- Wireshark
- Nmap
- Metasploitable Virtual Machine

Testbed Setup

Step 1: Start Metasploitable

1. Boot the Metasploitable virtual machine.
2. Determine and record its IP address using the following command:

```
ifconfig
```

or

```
ip addr
```

3. Note the IP address for later use.

Metasploitable IP Address: _____

Step 2: Install Required Tools

1. Download and install Nmap on the Windows machine.
2. Launch Wireshark.
3. Select the active LAN/Ethernet interface.
4. Start packet capture.

Part A: Generate Network Traffic Using Nmap

Open Command Prompt and execute the following commands, replacing <Metasploitable_IP> with the actual IP address of the target system.

Host Discovery and Scanning

```
nmap -A <Metasploitable_IP>
nmap -T1 <Metasploitable_IP>
nmap -T4 <Metasploitable_IP>
nmap -T5 <Metasploitable_IP>
nmap -sn 192.168.1.0/24
```

TCP Scan Types

```
nmap -sS <Metasploitable_IP>
nmap -sT <Metasploitable_IP>
```

Port Scanning

```
nmap -p 1-1000 <Metasploitable_IP>
nmap -sS -p- -T4 <Metasploitable_IP>
```

UDP and Service Detection

```
nmap -sU <Metasploitable_IP>
nmap -sV <Metasploitable_IP>
```

Operating System Detection

```
nmap -O <Metasploitable_IP>
```

Part B: Generate Web Traffic

Open a web browser and visit the following websites:

1. <http://www.cdac.in>
2. <http://httpforever.com>
3. <http://eitc.org>

Login Page Exploration

Use the following search queries to locate publicly accessible login pages:

```
inurl:http:// "login"  
inurl:http:// inurl:login  
inurl:http:// intitle:login
```

Browse at least five login pages discovered through these searches.

Important: Do not attempt unauthorized access to any website. Only observe page loading and associated network traffic. Do not submit credentials to systems you do not own or have permission to access.

Part C: Stop Packet Capture

After completing all Nmap scans and web browsing activities:

1. Return to Wireshark.
2. Stop the packet capture.
3. Save the capture file for analysis.

Part D: Apply Wireshark Display Filters

Display filters allow analysis of captured traffic without modifying the original capture.

Protocol Filters

Display Filter	Purpose
tcp	Show only TCP packets
udp	Show only UDP packets

icmp	Show only ICMP packets
dns	Show only DNS traffic
http	Show only HTTP traffic

IP Address Filters

Display Filter	Purpose
ip.addr == Your_IP	Traffic to or from your system
ip.src == 192.168.0.1	Packets from a specific source
ip.dst == <Metasploitable_IP>	Packets destined for Metasploitable

Port Filters

Display Filter	Purpose
tcp.port == 80	HTTP traffic
tcp.port == 443	HTTPS traffic

Packet Characteristics

Display Filter	Purpose
frame.len > 1000	Large packets
tcp.flags.syn == 1	TCP SYN packets
tcp.analysis.retransmission	TCP retransmissions

Part E: Combined Filters

Apply the following filters and observe the results.

```
tcp and ip.addr == 192.168.1.1
http or dns
tcp.port == 80 and ip.src == Your_IP
!(icmp)
```

Part F: TCP Flag Analysis

Display Filter	Purpose
tcp.flags.syn == 1	SYN packets
tcp.flags.ack == 1	ACK packets

<code>tcp.flags.fin == 1</code>	Connection termination
<code>tcp.flags.reset == 1</code>	TCP Reset packets
<code>tcp.flags.syn == 1 && tcp.flags.ack == 1</code>	SYN-ACK packets

Part G: DNS Traffic Analysis

Display Filter	Purpose
<code>dns.flags.response == 0</code>	DNS queries
<code>dns.flags.response == 1</code>	DNS responses
<code>dns.qry.name contains "google"</code>	Queries containing "google"
<code>dns.time > 1</code>	Slow DNS responses

Part H: IP Packet Analysis

Display Filter	Purpose
<code>ip.ttl < 10</code>	Low TTL packets
<code>ip.flags.df == 1</code>	Don't Fragment flag set
<code>ip.len > 1000</code>	Large IP packets
<code>ip.src == 192.168.1.10 && ip.dst == 8.8.8.8</code>	Specific communication path

Part I: String Matching Filters

Search for specific strings within captured packet data.

```
http contains "password"
frame contains "admin"
```

Record and analyze any observations.

Part J: Nmap Scan Identification

Use Wireshark to distinguish between different Nmap scan types.

SYN Scan

Filter:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Connect Scan

Filter:

```
tcp.flags.syn == 1 || tcp.flags.ack == 1
```

UDP Scan

Filter:

```
udp
```

OS Detection

Filter:

```
icmp or tcp.options
```

Tasks

1. Compare traffic generated by:
 - nmap -sS
 - nmap -sT
 - nmap -sU
2. Count the number of packets generated by each scan.
3. Identify which scan was fastest.
4. Determine how Wireshark can distinguish between scan types.

Part K: ARP Analysis

Generate ARP traffic by pinging another host.

Filter:

```
arp
```

Additional Filters:

```
arp.opcode == 1  
arp.opcode == 2
```

Tasks

1. Identify ARP Requests.
2. Identify ARP Replies.
3. Determine MAC addresses of communicating devices.
4. Explain ARP resolution.

Part L: ICMP Analysis

Generate ICMP traffic:

```
ping <Metasploitable_IP>
```

Filters:

```
icmp  
icmp.type == 8  
icmp.type == 0
```

Tasks

1. Identify Echo Requests.
2. Identify Echo Replies.
3. Measure round-trip time.
4. Determine packet size.

Part M: Security and Credential Discovery

Apply the following filters:

```
http contains "login"  
http contains "user"  
http contains "username"  
http contains "password"  
frame contains "admin"
```

Tasks

1. Search for login-related traffic.
2. Identify web forms transmitted over HTTP.
3. Discuss why plaintext HTTP is insecure.
4. Explain how HTTPS protects credentials.

Note: Only analyze traffic generated in the laboratory environment and systems for which you have authorization.

Part N: Statistics and Traffic Profiling

Use the following Wireshark menus:

Protocol Hierarchy

Statistics → Protocol Hierarchy

Record:

- Total packets captured
- TCP packet count

- UDP packet count
- DNS packet count
- HTTP packet count

Conversations

Statistics → Conversations

Record:

- Top source IPs
- Top destination IPs
- Most active TCP conversations

Endpoints

Statistics → Endpoints

Record:

- Number of hosts discovered
- Most active host

Part O: Packet Reconstruction

Follow TCP Stream

Right-click a TCP packet:

Follow → TCP Stream

Tasks

1. Reconstruct a complete HTTP session.
2. Identify request and response headers.
3. Record the server banner.
4. Identify webpage contents returned by the server.

Part P: Mini Network Forensics Challenge

Answer the following questions using Wireshark:

1. What is the IP address of the Metasploitable machine?
2. What operating system was detected by Nmap?
3. Which ports were open?
4. Which DNS server was used?
5. Which websites were visited?
6. Which host generated the most traffic?

7. Was any plaintext login information observed?
8. Which Nmap scan generated the highest packet count?
9. How many TCP conversations were established?
10. Which protocol generated the most traffic?

Deliverables

Submit a lab report containing:

1. Metasploitable IP address used.
2. Screenshots of:
 - Wireshark capture window
 - Nmap scan results
 - Applied display filters
3. Analysis of at least five display filters.
4. Identification of:
 - TCP SYN packets
 - DNS queries and responses
 - HTTP requests
 - Nmap scanning behavior
5. Conclusions summarizing key observations and lessons learned.
6. Screenshots of at least 15 display filters.
7. TCP three-way handshake analysis.
8. DNS resolution analysis.
9. ARP request/reply analysis.
10. HTTP session reconstruction.
11. Nmap scan comparison table.